

Large Language Models: Prompting

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Introduction

Introduction

연세대학교에 대해 알려줘



연세대학교(Yonsei University)는 1885년에 설립된 대한민국의 대표적인 사립 종합대학교로, 서울특별시 서대문구 신촌동에 본교를 두고 있습니다. 서울대학교, 고려대학교와 함께 'SKY'로 불리며 국내에서 가장 명망 있는 대학 중 하나로 평가받고 있습니다. [위키백과](#)

Introduction

- Language models can solve computer tasks (Kim et al., 2023)

Use the terminal below to list all files and delete a file ending with the extension .jpg

```
terminal
Welcome! Type help for a list
of available commands.
Last login: Thu Mar 16 2023
user$
```

For the user **@kenda**, click on the "Like" button.

Use the textbox to enter "**Dannie**" and press "Search", then find and click the **4th** search result.

Playing as 'X', win a game of tic-tac-toe.

Expand the sections below, to find and click on the link "vitae".

Section #1
Section #2
Section #3

Guess the number between 0-9 and press Submit. Use the feedback below to find the right number.

Click on the colored box.

Click on the "no" button.

Find the email by **Brier** and click the trash icon to delete it.

Switch between the tabs to find and click on the link "Quis".

Waiting for your guess...

Submit

vitae in morbi pellentesque mauris diam
no
sagittis, dolor, accumsan ultricies nibh facilisis
submit

Primary
Kiersten Pellentesque tr.. Mi ullamcorper ..
Marin Neque. Magna elementum..
Fredia Imperdiet. Nulla nunc cons..

Tab #1 Tab #2 Tab #3

Rhonus rutrum amet, habitant. Sagittis [gravida](#) sed ornare potenti. [Et, id blandit](#) id malesuada purus praesent egestas orci. [Quis purus](#).

Figure 1: MiniWoB++ environment. Every task contains a natural language prompt in yellow. The agent then uses keyboard strokes and mouse clicks to accomplish the task.

Introduction

- LLMs can also automate web navigation (Zhou et al., 2023)

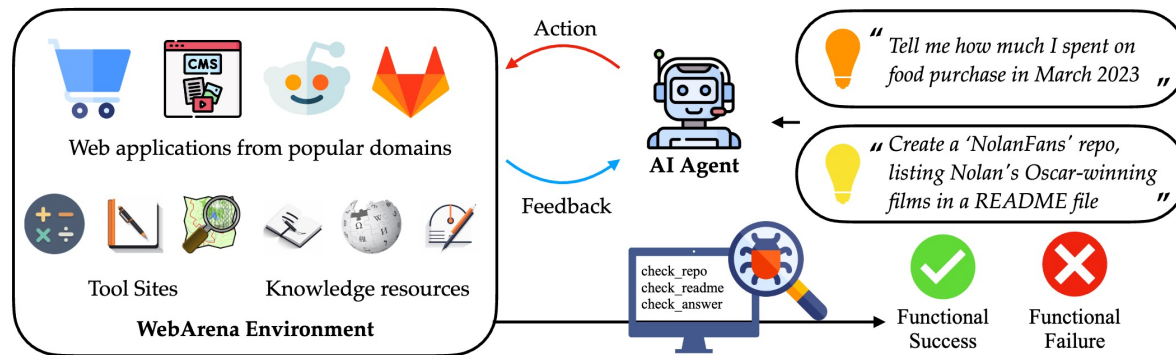


Figure 1: WebArena is a standalone, self-hostable web environment for building autonomous agents. WebArena creates websites from four popular categories with functionality and data mimicking their real-world equivalents. To emulate human problem-solving, WebArena also embeds tools and knowledge resources as independent websites. WebArena introduces a benchmark on interpreting *high-level realistic* natural language command to concrete web-based interactions. We provide validators to programmatically validate the functional correctness of each task.

Introduction

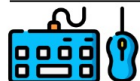
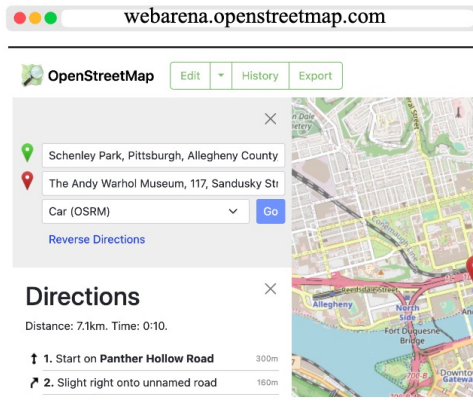
- LLMs can also automate web navigation (Zhou et al., 2023)



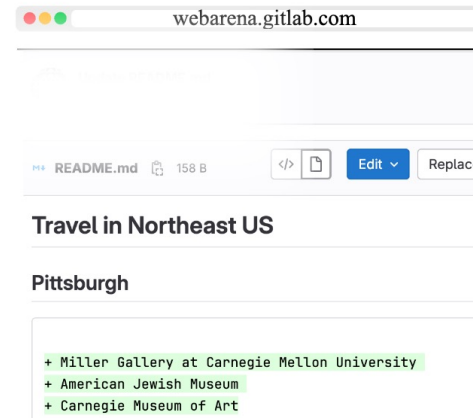
“ Create an efficient itinerary to visit all of Pittsburgh's art museums with minimal driving distance starting from Schenley Park. Log the order in my “awesome-northeast-us-travel” repository ”



Search for museums
in Pittsburgh



Search for each art
museum on the Map



Record the optimized
results to the repo

Introduction

- LLMs can also be clinical reasoners (Kwon et al., 2023)

Large Language Models Are Clinical Reasoners: Reasoning-Aware Diagnosis Framework with Prompt-Generated Rationales

**Taeyoon Kwon^{1*}, Kai Tzu-iunn Ong^{1*}, Dongjin Kang², Seungjun Moon², Jeong Ryong Lee³,
Dosik Hwang³, Beomseok Sohn⁴, Yongsik Sim⁴, Dongha Lee¹, Jinyoung Yeo¹**

¹Department of Artificial Intelligence, Yonsei University

²Department of Computer Science and Engineering, Yonsei University

³Department of Electrical and Electronic Engineering, Yonsei University

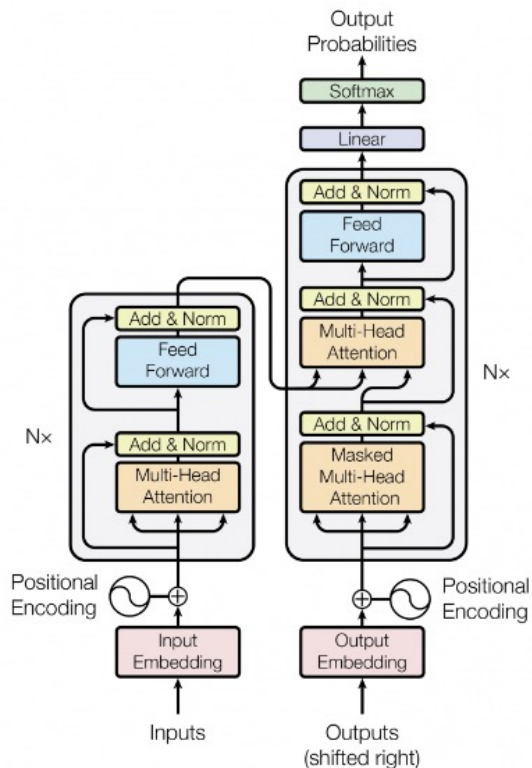
⁴Department of Radiology, College of Medicine, Yonsei University

Introduction

- Let's see how this progress become possible within just a few years!!!

Recap

Recap – Transformer



- **Attention is All you Need (2017)**

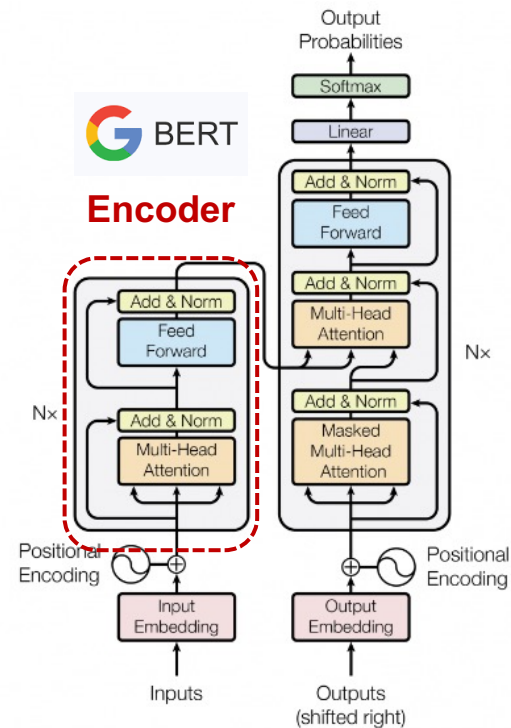
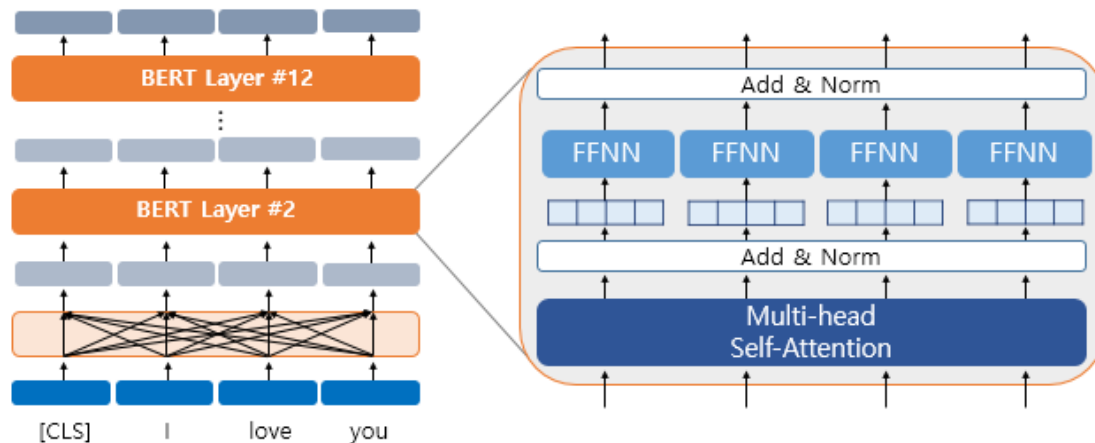
- **Why Transformer?**

- Self-Attention
- No recurrence
- Scalability
- Parallelizability

...

➔ **The foundation of nearly all modern DL models**

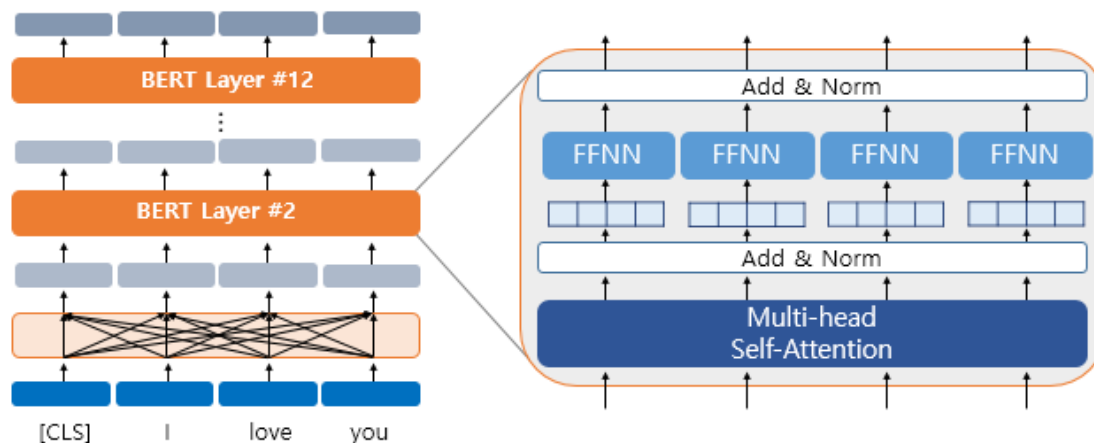
Recap – BERT



Recap – BERT

- **BERT is a bidirectional encoder model**

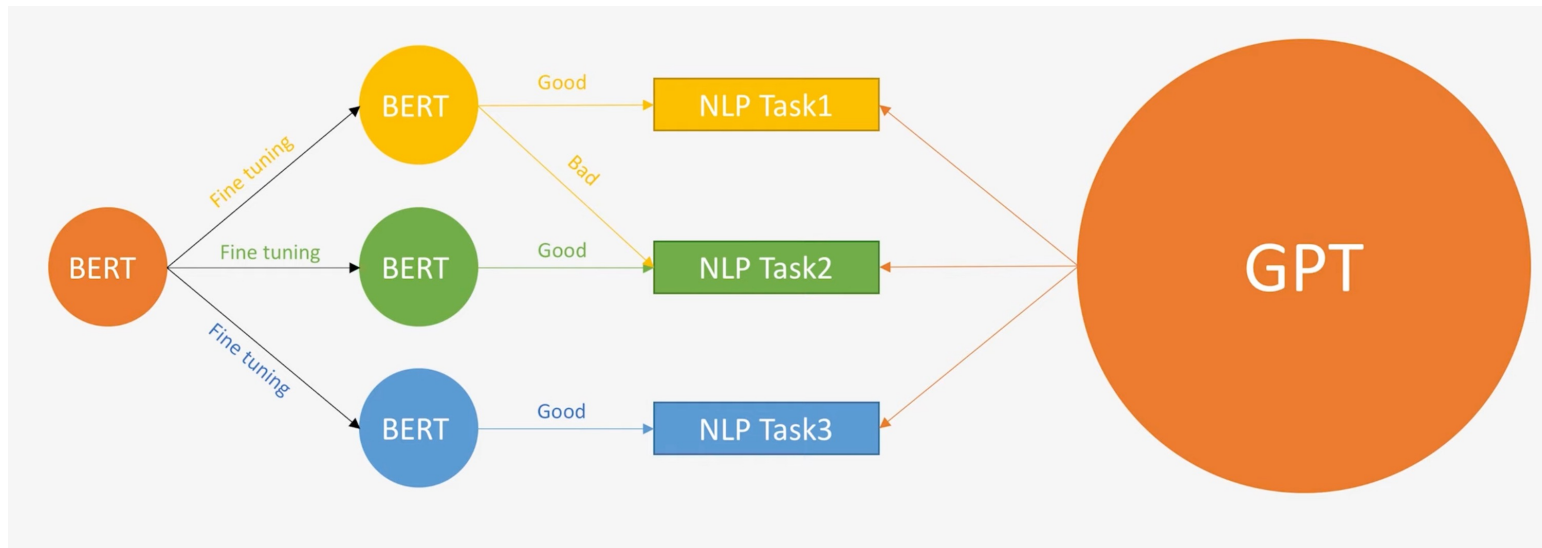
- Trained with **two objectives**:
 - **Masked Language Modeling (MLM)**
 - **Next Sentence Prediction (NSP)**
- Excels at **natural language understanding tasks** with full sentence context
- Fine-tuned for each downstream task (e.g., QA, classification, NER)



Recap – GPT

- **GPT is a general-purpose model**

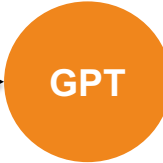
- Trained with **single objective**: **next-token prediction**
- Performs well on many tasks with **zero-shot** or **few-shot prompting**



Recap – GPT

- **Input = Task + Context**
 - Output → predict the most probable next token
- **GPT-2's improved performance enables it to handle multiple tasks without fine-tuning — using just prompts.**

$$p(\text{output}|\text{input}) \longrightarrow p(\text{output}|\text{input}, \text{task})$$

“Brevet Sans Garantie Du Gouvernement”, translated to English:  “Patented without government warranty”.

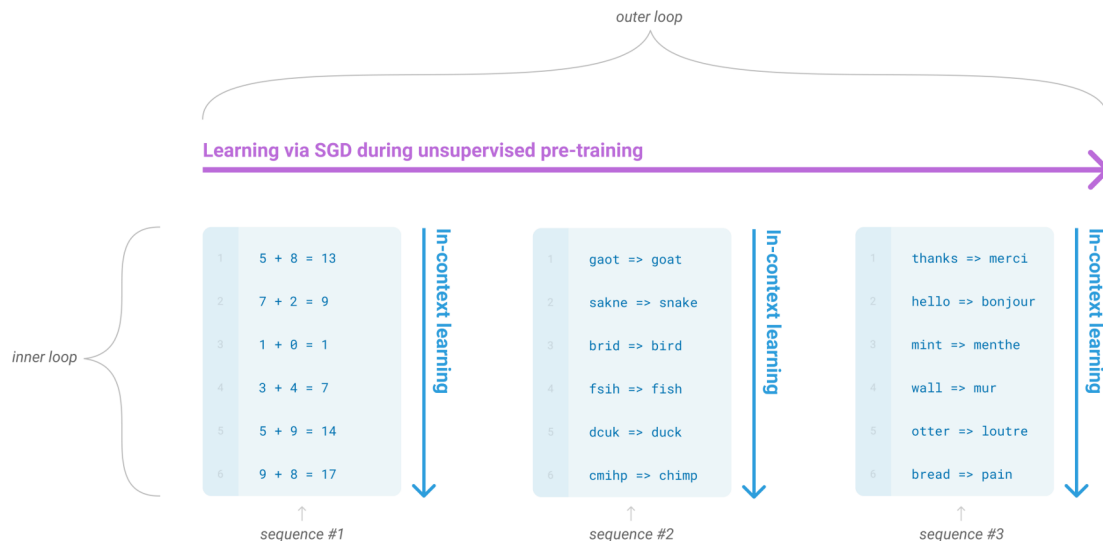
Emergent of Large Language Models

In-Context Learning

In-Context Learning

- **Task Learning via Context, not Parameters**

- Language models can learn from few-shot demonstrations (**Without parameter update**)
- Previous: Fine-tuning (**Difficult and expensive** to collect training datasets)
- Emergent phenomenon in LLMs



In-Context Learning

- Learning “how to learn”

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



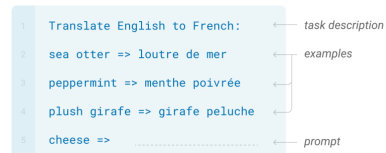
One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



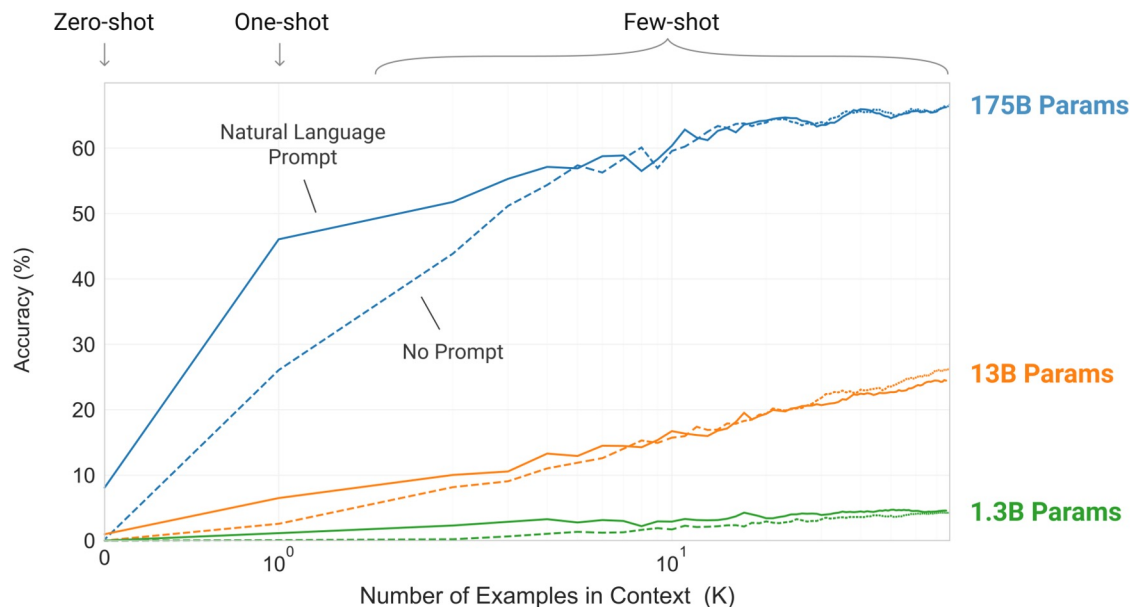
Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



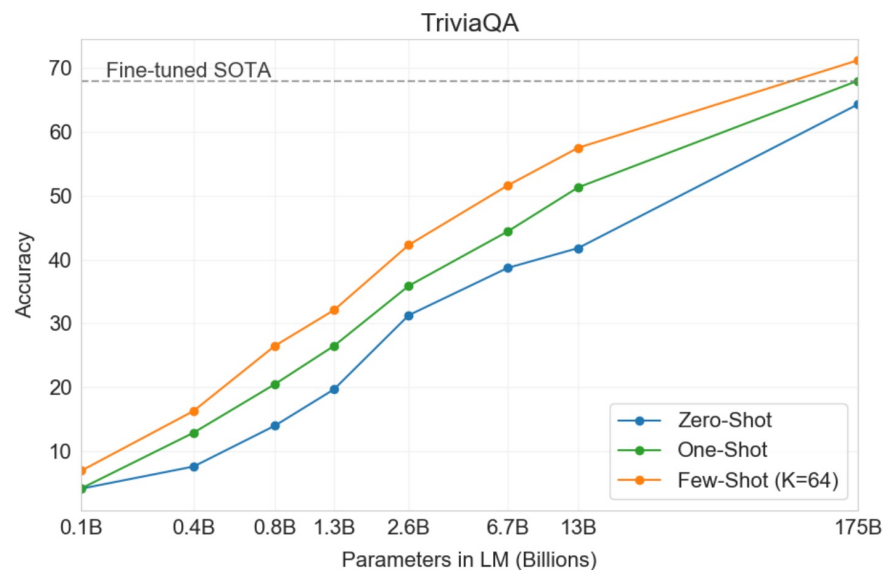
In-Context Learning

- The larger the model is, the better the accuracy is



In-Context Learning

- Few-shot > One-shot > Zero-shot



Task 1-1: In-Context Learning

- Task: Mathematical Reasoning (GSM8K)

 **Hugging Face**

Models Datasets S

Datasets:  **openai/gsm8k**  like 291

Tasks:  Text2Text Generation Modalities:  Text Formats:  parquet Languages:  English Size: 10K - 100K ArXiv:  arxiv:2110.14168

Libraries:  Datasets  pandas  Croissant License:  mit

 **Dataset card**  Viewer  Files and versions  Community 6

Dataset Viewer  Auto-converted to Parquet  API  View in Dataset Viewer

Subset (2)	Split (2)
main · 8.79k rows	train · 7.47k rows

question	answer
string · lengths	string · lengths
	
Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell...	Natalia sold $48/2 = <<48/2=24>>24$ clips in May. Natalia sold $48+24 = <<48+24=72>>72$ clips altogether in April and May. ##### 72
Weng earns \$12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?	Weng earns $12/60 = \$<<12/60=0.2>>0.2$ per minute. Working 50 minutes, she earned $0.2 \times 50 = \$<<0.2 \times 50=10>>10$. ##### 10
Betty is saving money for a new wallet which costs \$100. Betty has only half of the money she needs. Her parents decided to give her...	In the beginning, Betty has only $100 / 2 = \$<<100/2=50>>50$. Betty's grandparents gave her $15 \times 2 = \$<<15 \times 2=30>>30$. This means...
Julie is reading a 120-page book. Yesterday, she was able to read 12 pages and today, she read twice as many pages as yesterday. If...	Maila read $12 \times 2 = <<12 \times 2=24>>24$ pages today. So she was able to read a total of $12 + 24 = <<12+24=36>>36$ pages since yesterday...

Task 1-1: In-Context Learning

- Examples

Question: Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

Answer: Natalia sold $48/2 = <<48/2=24>>24$ clips in May. Natalia sold $48+24 = <<48+24=72>>72$ clips altogether in April and May. ##### 72

Task 1-1: In-Context Learning

- In-context demonstration construction

[Example 1]

Input:

~~~

Output:

~~~

...

[Example 4]

Input:

~~~

Output:



# Task 1-1: In-Context Learning

- In-context demonstration construction

## [Example 1]

Input:

~~~

Output:

~~~

...

## [Example 4]

Input:

~~~

Output: 

LLM will generate output after this identifier!

Grok API

- Through Grok API, you can get generation results of various chat models.

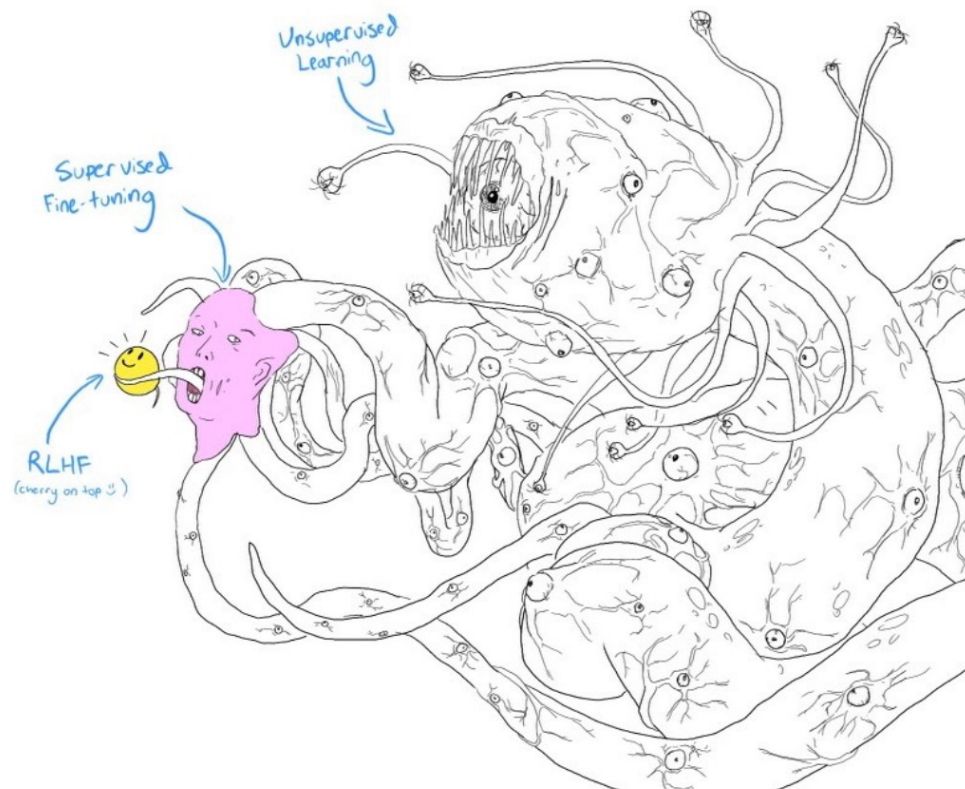
Instruction Tuning

Challenges of Pre-trained LLMs

- **Even LLMs suffer to understand human's intention and value**
 - LLM's are trained to predict next token
 - Low-quality data in web

Prompt	<i>Explain the moon landing to a 6 year old in a few sentences.</i>
Completion	GPT-3 Explain the theory of gravity to a 6 year old. Explain the theory of relativity to a 6 year old in a few sentences. Explain the big bang theory to a 6 year old. Explain evolution to a 6 year old. InstructGPT People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

Challenges of Pre-trained LLMs



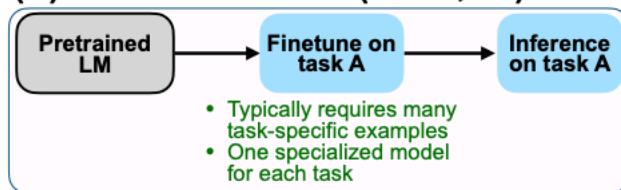
Instruction Tuning

- **Previous: Learning mapping from input to output without instruction**

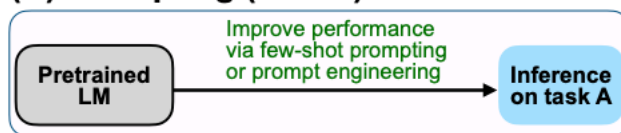
Instruction Tuning

- Previous: Learning mapping from input to output without instruction
- Now: Training LMs to follow instructions that describe the tasks asked for LMs to complete

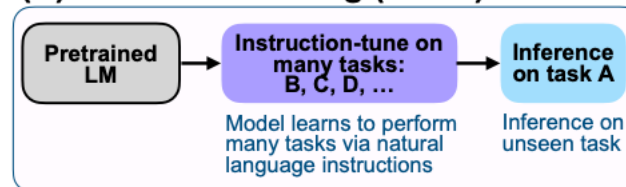
(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)



(C) Instruction tuning (FLAN)



Instruction Tuning

- Instruction following capabilities are generalized to unseen tasks

Resource →	SUP-NATINST (this work)	NATINST (Mishra et al., 2022b)	CROSSFIT (Ye et al., 2021)	PROMPTSOURCE (Bach et al., 2022)	FLAN (Wei et al., 2022)	INSTRUCTGPT (Ouyang et al., 2022)
Has task instructions?	✓	✓	✗	✓	✓	✓
Has negative examples?	✓	✓	✗	✗	✗	✗
Has non-English tasks?	✓	✗	✗	✗	✓	✓
Is public?	✓	✓	✓	✓	✓	✗
Number of tasks	1616	61	269	176	62	—
Number of instructions	1616	61	—	2052	620	14378
Number of annotated tasks types	76	6	13	13*	12	10
Avg. task definition length (words)	56.6	134.4	—	24.8	8.2	—



Task 1-2: Instruction following LMs

- Task: Add instruction and also see the zero-shot performance!

Instruction: ~~~

[Example 1]

Input:

~~~

Output:

~~~

...

[Example 4]

Input:

~~~

Output:

# Task 1-2: Instruction following LMs

- Task: Add instruction and also see the zero-shot performance!

### Instruction: ~~~

[Example 1]

Input:

~~~

Output:



Zero shot!

How can we better use LLMs?

Chain-of-Thought Prompting

Chain-of-Thought Prompting

- LLMs can improve their reasoning capabilities through intermediate natural language reasoning steps before generating answer
 - Emergent phenomenon in LLMs

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

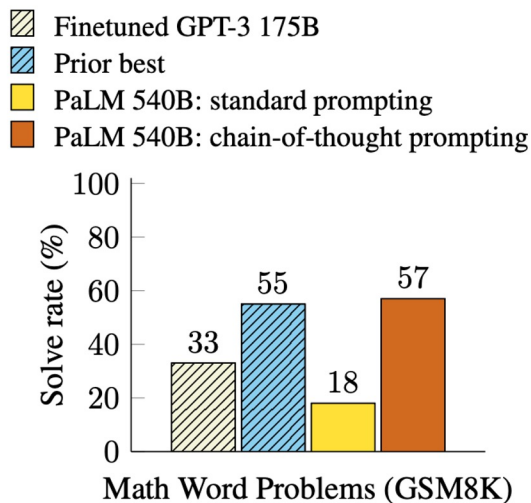
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

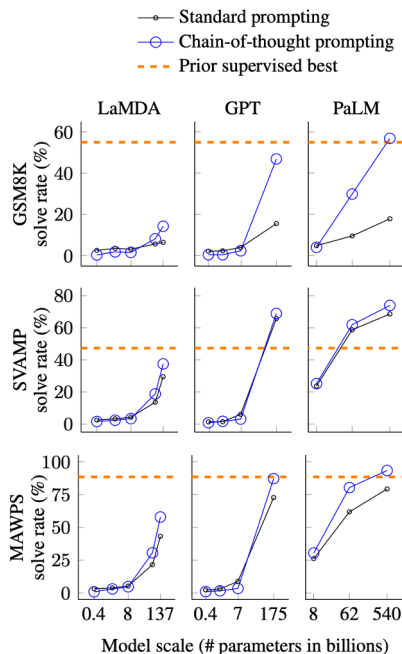
Chain-of-Thought Prompting

- **CoT outperforms standard prompting by a large margin and achieves new state-of-the-art performance.**
 - Even better performance than fine-tuned LLMs.



Chain-of-Thought Prompting

- **CoT is an emergent phenomenon of the LLMs**
 - Model scale



Chain-of-Thought Prompting

- How to generate intermediate reasoning steps?
 - Few shot demonstrations!!!

PROMPT FOR AQUA ALGEBRAIC WORD PROBLEMS

Q: John found that the average of 15 numbers is 40. If 10 is added to each number then the mean of the numbers is?

Answer Choices: (a) 50 (b) 45 (c) 65 (d) 78 (e) 64

A: If 10 is added to each number, then the mean of the numbers also increases by 10. So the new mean would be 50. The answer is (a).

Q: If $a/b = 3/4$ and $8a + 5b = 22$, then find the value of a .

Answer Choices: (a) $1/2$ (b) $3/2$ (c) $5/2$ (d) $4/2$ (e) $7/2$

A: If $a/b = 3/4$, then $b = 4a/3$. So $8a + 5(4a/3) = 22$. This simplifies to $8a + 20a/3 = 22$, which means $44a/3 = 22$. So a is equal to $3/2$. The answer is (b).

Q: A person is traveling at 20 km/hr and reached his destiny in 2.5 hr then find the distance?

Answer Choices: (a) 53 km (b) 55 km (c) 52 km (d) 60 km (e) 50 km

A: The distance that the person traveled would have been $20 \text{ km/hr} * 2.5 \text{ hrs} = 50 \text{ km}$. The answer is (e).

Q: How many keystrokes are needed to type the numbers from 1 to 500?

Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$. The answer is (b).

Task 2-1: CoT Prompting

- **Solve GSM8K with Chain-of-Thought!!**
 - Better with instruction??

[Example 1]

Input:

~~~

Output:

In the beginning, Betty has only  $100 / 2 = \$\langle\langle 100/2=50 \rangle\rangle 50$ .

Betty's grandparents gave her  $15 * 2 = \$\langle\langle 15*2=30 \rangle\rangle 30$ .

This means, Betty needs  $100 - 50 - 30 - 15 = \$\langle\langle 100-50-30-15=5 \rangle\rangle 5$  more.

#### 5

...

## [Example 4]

Input:

~~~

Output:

Task 2-1: CoT Prompting

- **How about Zero-shot CoT??**
 - Add the instruction?

Instruction: Generate the answer with rationale~~~

[Example 1]

Input:

~~~

Output:

# **Zero-shot Chain-of-Thought Prompting**

# Zero-shot CoT

- How can we generate reasoning steps without any demonstrations?

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are  $16 / 2 = 8$  golf balls. Half of the golf balls are blue. So there are  $8 / 2 = 4$  blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

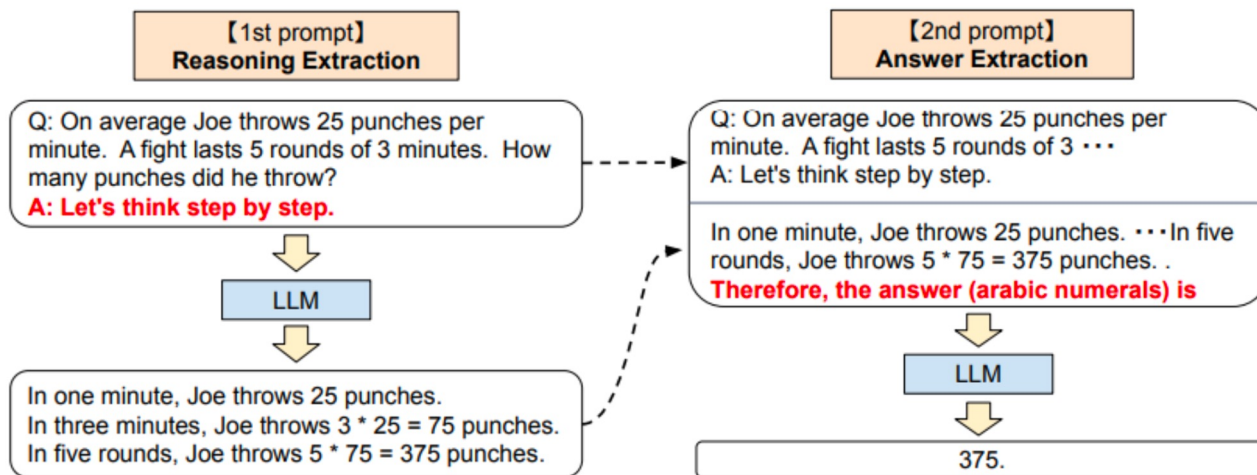
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

# Zero-shot CoT

- Two stage prompting
  - Reasoning extraction and answer extraction



# Zero-shot CoT

- **Two stage prompting**
  - Reasoning extraction and answer extraction

|                                                                | MultiArith  | GSM8K       |
|----------------------------------------------------------------|-------------|-------------|
| <b>Zero-Shot</b>                                               | <b>17.7</b> | <b>10.4</b> |
| Few-Shot (2 samples)                                           | 33.7        | 15.6        |
| Few-Shot (8 samples)                                           | 33.8        | 15.6        |
| <b>Zero-Shot-CoT</b>                                           | <b>78.7</b> | <b>40.7</b> |
| Few-Shot-CoT (2 samples)                                       | 84.8        | 41.3        |
| Few-Shot-CoT (4 samples : First) (*1)                          | 89.2        | -           |
| Few-Shot-CoT (4 samples : Second) (*1)                         | 90.5        | -           |
| Few-Shot-CoT (8 samples)                                       | 93.0        | 48.7        |
| <b>Zero-Plus-Few-Shot-CoT (8 samples) (*2)</b>                 | <b>92.8</b> | <b>51.5</b> |
| Finetuned GPT-3 175B [Wei et al., 2022]                        | -           | 33          |
| Finetuned GPT-3 175B + verifier [Wei et al., 2022]             | -           | 55          |
| <b>PaLM 540B: Zero-Shot</b>                                    | <b>25.5</b> | <b>12.5</b> |
| <b>PaLM 540B: Zero-Shot-CoT</b>                                | <b>66.1</b> | <b>43.0</b> |
| <b>PaLM 540B: Zero-Shot-CoT + self consistency</b>             | <b>89.0</b> | <b>70.1</b> |
| PaLM 540B: Few-Shot [Wei et al., 2022]                         | -           | 17.9        |
| PaLM 540B: Few-Shot-CoT [Wei et al., 2022]                     | -           | 56.9        |
| PaLM 540B: Few-Shot-CoT + self consistency [Wang et al., 2022] | -           | 74.4        |

# Task 2-1: CoT Prompting

- Zero-shot CoT with reasoning extraction!

## [Example 1]

Input:

~~~

Output:

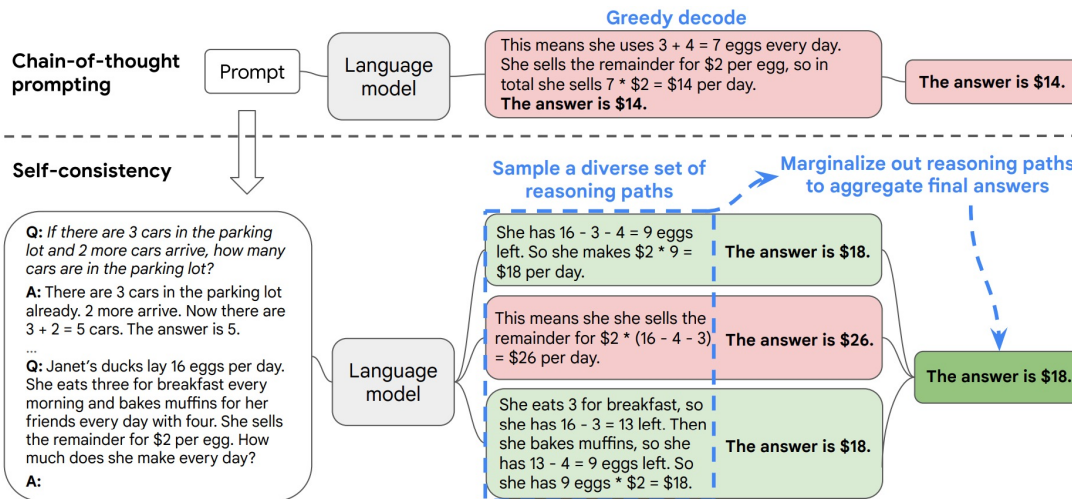
[Answer]

Let's think step by step.

Self-Consistency

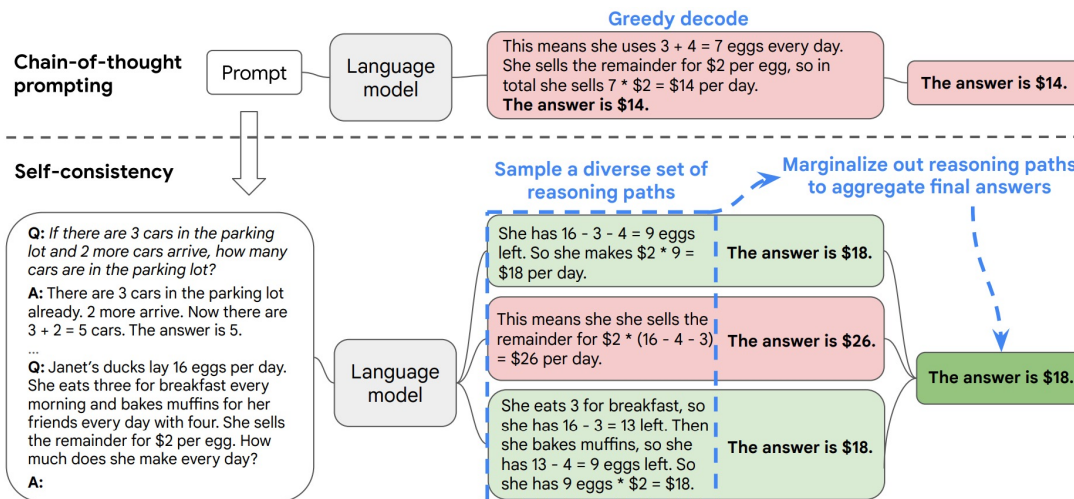
Self-Consistency

- CoT decoding strategy: Greedy decoding



Self-Consistency

- **Complex reasoning tasks admit multiple reasoning paths**
 - The more deliberate thinking is required, the greater the diversity of reasoning path is.



Self-Consistency

- **Self-consistency significantly improves the reasoning performance**
 - Mathematical reasoning

	Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
	Previous SoTA	94.9^a	60.5 ^a	75.3 ^b	37.9 ^c	57.4 ^d	35 ^e / 55 ^g
UL2-20B	CoT-prompting	18.2	10.7	16.9	23.6	12.6	4.1
	Self-consistency	24.8 (+6.6)	15.0 (+4.3)	21.5 (+4.6)	26.9 (+3.3)	19.4 (+6.8)	7.3 (+3.2)
LaMDA-137B	CoT-prompting	52.9	51.8	49.0	17.7	38.9	17.1
	Self-consistency	63.5 (+10.6)	75.7 (+23.9)	58.2 (+9.2)	26.8 (+9.1)	53.3 (+14.4)	27.7 (+10.6)
PaLM-540B	CoT-prompting	91.9	94.7	74.0	35.8	79.0	56.5
	Self-consistency	93.7 (+1.8)	99.3 (+4.6)	81.9 (+7.9)	48.3 (+12.5)	86.6 (+7.6)	74.4 (+17.9)
GPT-3 Code-davinci-001	CoT-prompting	57.2	59.5	52.7	18.9	39.8	14.6
	Self-consistency	67.8 (+10.6)	82.7 (+23.2)	61.9 (+9.2)	25.6 (+6.7)	54.5 (+14.7)	23.4 (+8.8)
GPT-3 Code-davinci-002	CoT-prompting	89.4	96.2	80.1	39.8	75.8	60.1
	Self-consistency	91.6 (+2.2)	100.0 (+3.8)	87.8 (+7.6)	52.0 (+12.2)	86.8 (+11.0)	78.0 (+17.9)

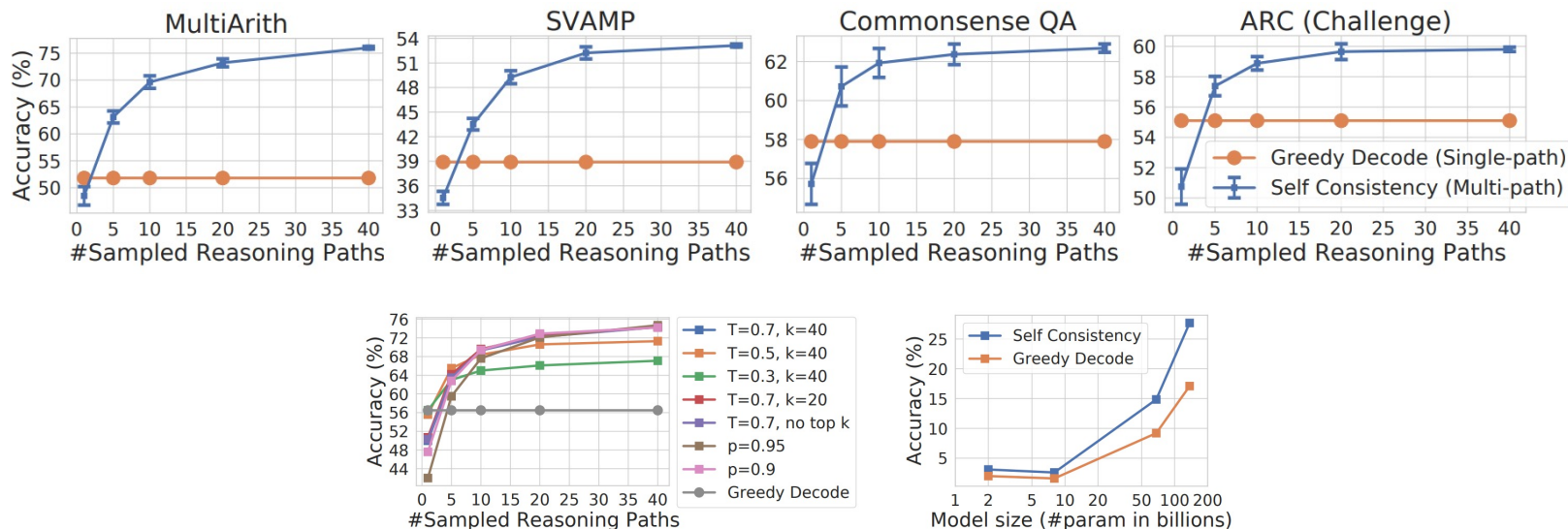
Self-Consistency

- **Self-consistency significantly improves the reasoning performance**
 - Commonsense reasoning

	Method	CSQA	StrategyQA	ARC-e	ARC-c	Letter (4)	Coinflip (4)
	Previous SoTA	91.2^a	73.9 ^b	86.4 ^c	75.0 ^c	N/A	N/A
UL2-20B	CoT-prompting	51.4	53.3	61.6	42.9	0.0	50.4
	Self-consistency	55.7 (+4.3)	54.9 (+1.6)	69.8 (+8.2)	49.5 (+6.8)	0.0 (+0.0)	50.5 (+0.1)
LaMDA-137B	CoT-prompting	57.9	65.4	75.3	55.1	8.2	72.4
	Self-consistency	63.1 (+5.2)	67.8 (+2.4)	79.3 (+4.0)	59.8 (+4.7)	8.2 (+0.0)	73.5 (+1.1)
PaLM-540B	CoT-prompting	79.0	75.3	95.3	85.2	65.8	88.2
	Self-consistency	80.7 (+1.7)	81.6 (+6.3)	96.4 (+1.1)	88.7 (+3.5)	70.8 (+5.0)	91.2 (+3.0)
GPT-3 Code-davinci-001	CoT-prompting	46.6	56.7	63.1	43.1	7.8	71.4
	Self-consistency	54.9 (+8.3)	61.7 (+5.0)	72.1 (+9.0)	53.7 (+10.6)	10.0 (+2.2)	75.9 (+4.5)
GPT-3 Code-davinci-002	CoT-prompting	79.0	73.4	94.0	83.6	70.4	99.0
	Self-consistency	81.5 (+2.5)	79.8 (+6.4)	96.0 (+2.0)	87.5 (+3.9)	73.4 (+3.0)	99.5 (+0.5)

Self-Consistency

- **Self-consistency significantly improves the reasoning performance**
 - Robust to the number of reasoning paths, sampling strategy and scaling



Task 2-2: Self-consistency

- **Let's apply self-consistency!**
 - How? Sample the examples multiple times!

[Example 1]

Input:

~~~

Output:

~~~

...

[Example 4]

Input:

~~~

Output:

## [Example 1\*]

Input:

~~~

Output:

~~~

...

## [Example 4\*]

Input:

~~~

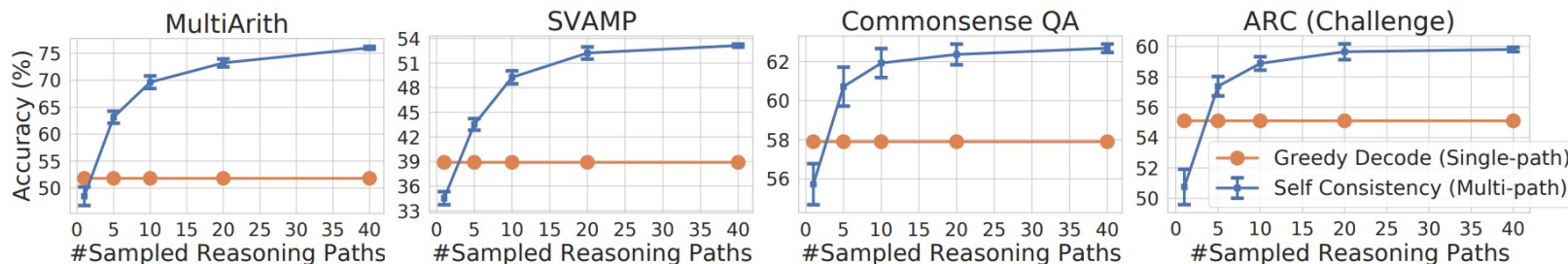
Output:

...

Task 2-2: Self-consistency

- Let's plot the result

- 1~5 times

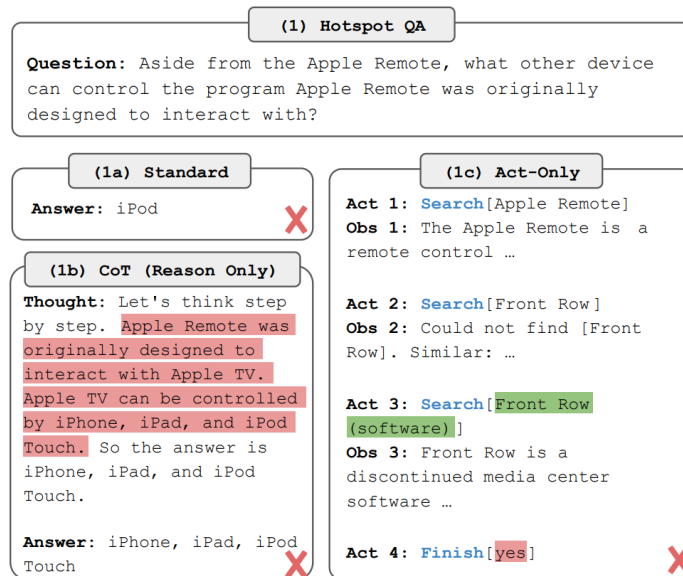


ReAct

ReAct

Recent approaches

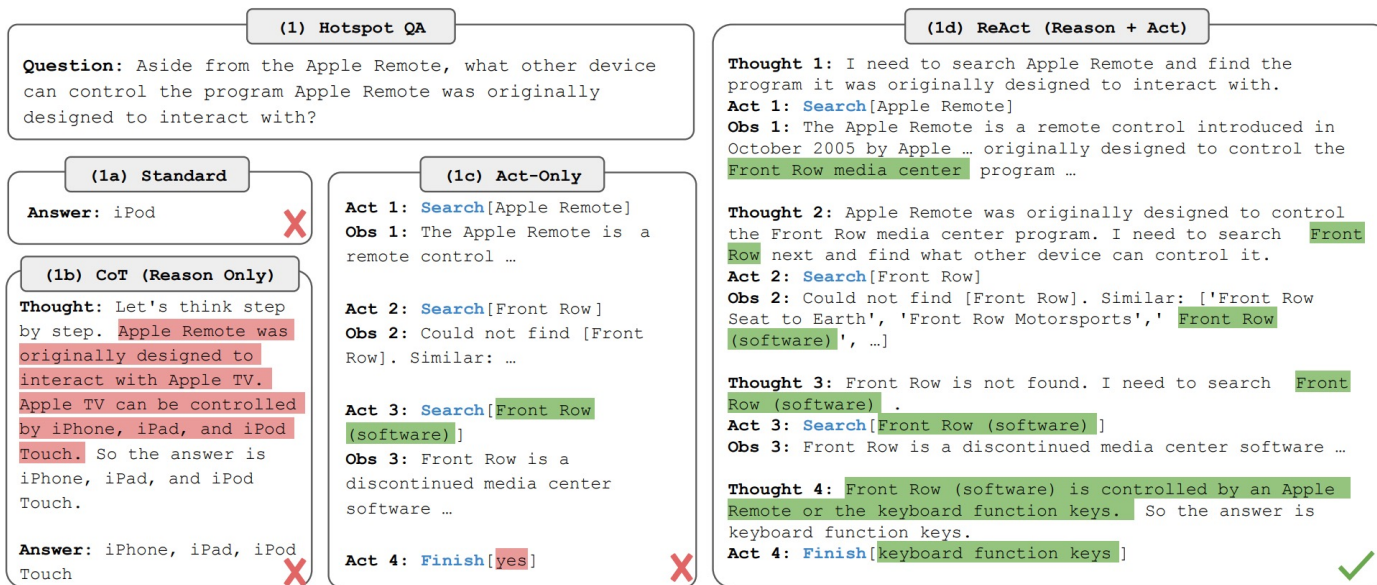
- Reasoning only: Internal representation only, not grounded in external world
-> hallucination, error propagation
- Actions: generate domain-specific actions or plans, and then use a controller to choose or execute them -> No reasoning abstractly about high-level goals



ReAct

Reasoning + Acting

- Prompt LLMs to generate both verbal reasoning traces and actions pertaining to a task



(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1a) Standard

Answer: iPod



(1b) CoT (Reason Only)

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch



(1c) Act-Only

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control ...

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ...

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Act 4: Finish[yes]



(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search Front Row (software) .

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: Finish[keyboard function keys]



ReAct

● Results

- HotpotQA: CoT outperforms
- Fever: Fact verification -> ReAct outperforms
- SC: Self-consistency

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
Supervised SoTA^b	67.5	89.5

ReAct

- **ReAct significantly mitigates the hallucination issue of CoT**

	Type	Definition	ReAct	CoT
Success	True positive	Correct reasoning trace and facts	94%	86%
	False positive	Hallucinated reasoning trace or facts	6%	14%
Failure	Reasoning error	Wrong reasoning trace (including failing to recover from repetitive steps)	47%	16%
	Search result error	Search return empty or does not contain useful information	23%	-
	Hallucination	Hallucinated reasoning trace or facts	0%	56%
	Label ambiguity	Right prediction but did not match the label precisely	29%	28%

Task 2-3: ReAct

- Task: Wikipedia-based question answering task
 - HotpotQA



Dataset Card for "hotpot_qa"

Dataset Summary

HotpotQA is a new dataset with 113k Wikipedia-based question-answer pairs with four key features: (1) the questions require finding and reasoning over multiple supporting documents to answer; (2) the questions are diverse and not constrained to any pre-existing knowledge bases or knowledge schemas; (3) we provide sentence-level supporting facts required for reasoning, allowing QA systems to reason with strong supervision and explain the predictions; (4) we offer a new type of factoid comparison questions to test QA systems' ability to extract relevant facts and perform necessary comparison.

Task 2-3: ReAct

- What to do
 - Let's compare the performance of CoT / Act / ReAct

(1) Hotspot QA

Question: How many rooms are in the hotel that is home to the Cirque du Soleil show Mystere?

HotspotQA label: 2,664

Outdated X

(1a) Standard

Answer: 3,000

X

(1b) CoT (Reason Only)

Thought: Let's think step by step. The hotel that is home to the Cirque du Soleil show Mystere is Treasure Island. Treasure Island has 2,885 rooms, so the answer is 2,885.

Answer: 2,885

X

(1c) Act-Only

Act 1: Search[Cirque du Soleil]
Obs 1: Cirque du Soleil is a Canadian entertainment company...

Act 2: Lookup[Mystere]
Obs 2: No more results

Act 3: Search[Mystere (Cirque du Soleil show)]
Obs 3: Could not find...

Act 4: Search[Mystere (Cirque du Soleil)]
Obs 4: Mystère (my-steer) is one of six...

Act 5: Lookup[Treasure Island Hotel and Casino]
(end without answer)

X

(1d) ReAct (Reason + Act)

Thought 1: I need to search Cirque du Soleil show Mystere, find the hotel it is in, then find the number of rooms in the hotel.

Act 1: Search[Cirque du Soleil show Mystere]
Obs 1: Could not find Cirque du Soleil show Mystere. Similar: ['Cirque du Soleil' ...]

Thought 2: To find the hotel, I can search Mystere (Cirque du Soleil).

Act 2: Search[Mystere (Cirque du Soleil)]
Obs 2: Mystère (my-steer) is ... Held in a custom theatre at the Treasure Island Hotel and Casino ...

Thought 3: Mystere is held in a custom theatre at the Treasure Island Hotel and Casino. So I need to search Treasure Island Hotel and Casino next and find the number of rooms in the hotel.

Act 3: Search[Treasure Island Hotel and Casino]
Obs 3: Treasure Island Hotel and Casino ... with 2,884 rooms and 220 suites ...

Thought 4: Treasure Island Hotel and Casino has 2,884 rooms and 220 suites. So the answer is 3,104.

Act 4: Finish[3,104]

Up-to-date ✓


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